

FAIR assessments

eLwazi ODSP 2026 Data Jamboree

Cape Town, 9-13 March 2026

Wei Kheng Teh, MSc

Senior Bioinformatician - BioSamples group
EMBL-EBI

<https://orcid.org/0000-0002-6583-8504>

wteh@ebi.ac.uk



Vassilios Ioannidis, PhD

Lead Computational Biologist - SIB/Vital-IT group
ELIXIR Switzerland

<https://orcid.org/0000-0002-4209-2578>

vassilios.ioannidis@sib.swiss

Danielle Welter, PhD

Principal Data Scientist - Luxembourg National Data Service (LNDS)
ELIXIR Luxembourg

<https://orcid.org/0000-0003-1058-2668>



www.elixir-europe.org



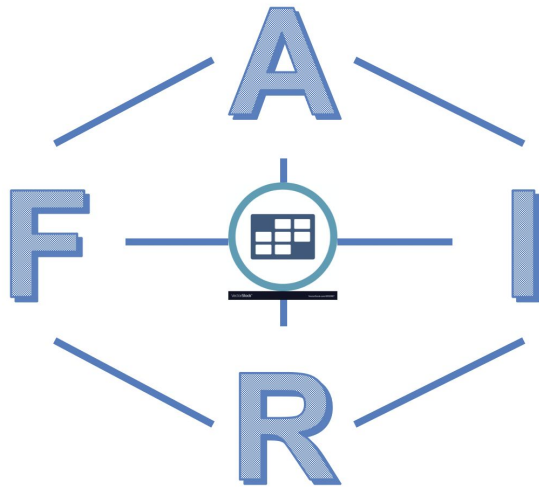
FAIRplus www.fairplus-project.eu

Repositories and databases

- ELIXIR Core Data Resources
<https://elixir-europe.org/platforms/data/core-data-resources>
- ELIXIR Deposition Databases for Biomolecular Data
<https://elixir-europe.org/platforms/data/elixir-deposition-databases>
- Global Biodata Coalition Core Biodata Resources
<https://globalbiodata.org/what-we-do/global-core-biodata-resources/list-of-current-global-core-biodata-resources/>

Assessing FAIR as a **state** vs FAIR as an Alphabet

- FA ≠ FAIR
- Very Good FA is still ≠ FAIR
- (Re)-**Use** is the goal NOT Findability, NOT Accessibility and NOT Interoperability



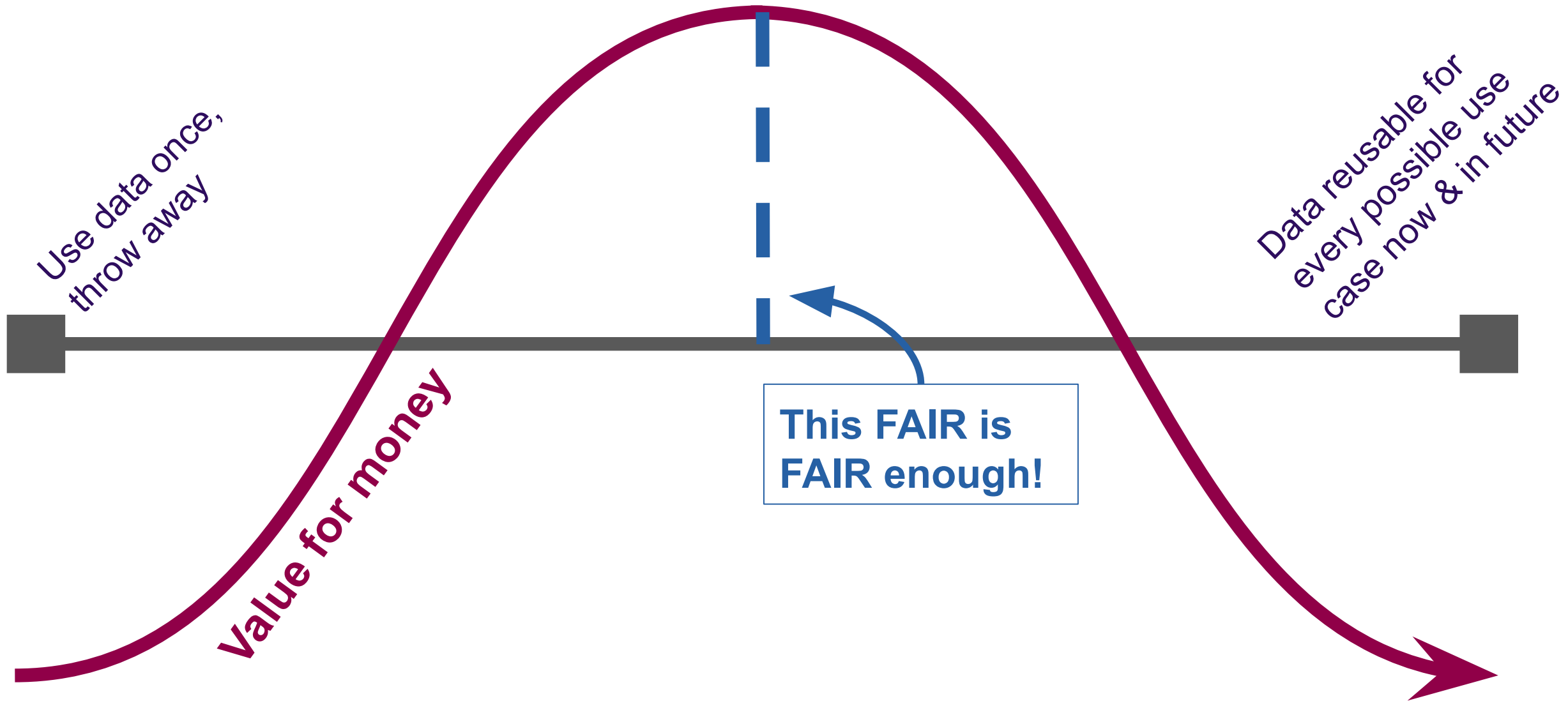
State of FAIRness

vs



FAIR score

HOW FAIR IS YOUR DATA? Is it FAIR enough?



Challenges with assessing FAIR

- Lots of different FAIR assessment tools & methodologies exist
- Results often vary wildly and are not comparable
- Different ways of conforming to a FAIR indicator
 - ◆ Your “FAIR enough” may not be the same as my “FAIR enough”



FAIRassist framework



FAIRassist
signposting tools

Discover **tools** for evaluation, assistance and assessment, or register your own

FAIRassist
principles, metrics, benchmarks

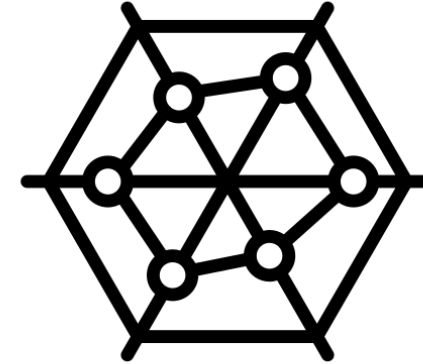
Discover **metrics** for evaluation, assistance and assessment tools, or register your own

<https://fairassist.org/>

FAIR as a checklist vs FAIR as a state



- ❑ Treat indicators in isolation
- ❑ Limited response options
- ❑ Useful for assessing resources (tools or services)
- ❑ Limited utility for different states of “FAIR enough”



- ❑ Dimensions of FAIRness
- ❑ Qualified responses
- ❑ Useful for assessing organisations, projects or datasets
- ❑ Dimensions and levels combine to define “FAIR enough”

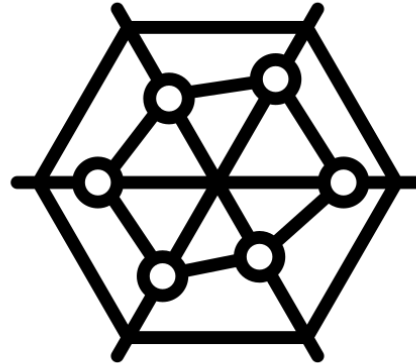
FAIR as a checklist examples

- ❑ RDA FAIR Data Maturity Model (<https://doi.org/10.15497/rda00050>)
- ❑ FAIR Checker (<https://fair-checker.france-bioinformatique.fr/>)
- ❑ FAIR Enough (<https://fair-enough.semanticscience.org/>)

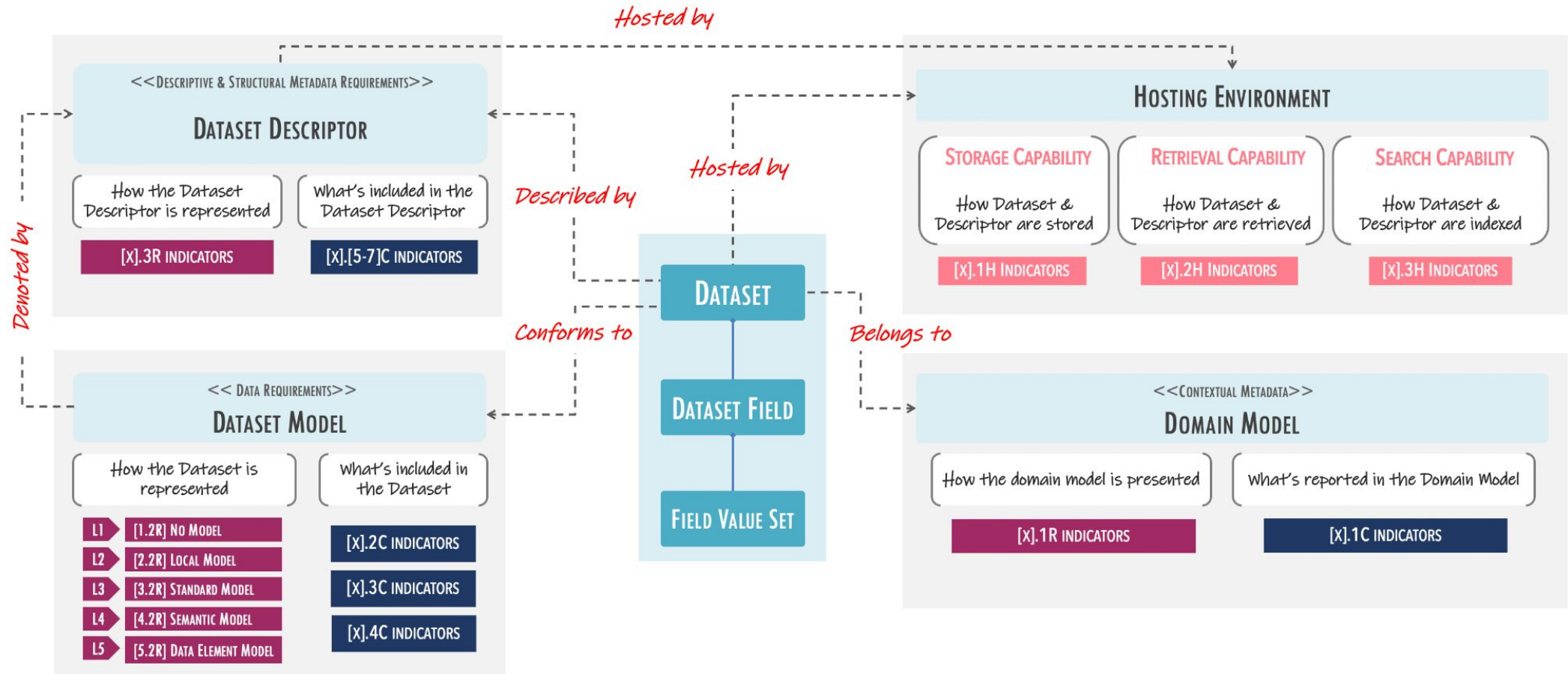


FAIR as a state examples

- ❑ FAIR DataSet Maturity (DSM) (<https://fairplus.github.io/Data-Maturity/>)
- ❑ “Do I-PASS for FAIR” (<https://doi.org/10.5281/zenodo.4080867>)
- ❑ **RDA FAIR Data Maturity Model* (<https://doi.org/10.15497/rda00050>)



FAIR-DSM Conceptual Model

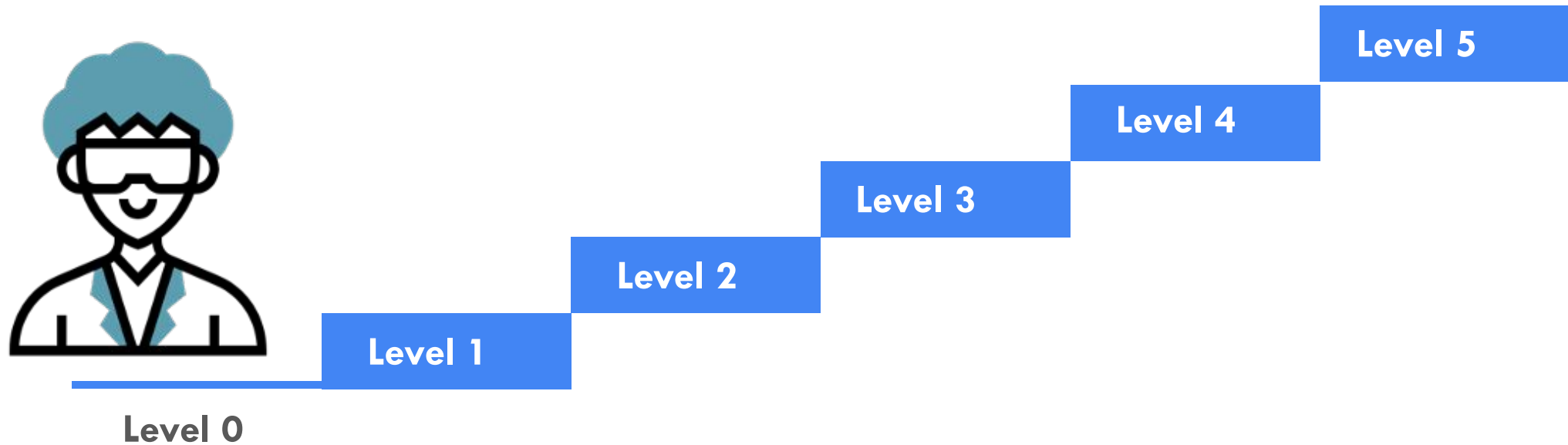


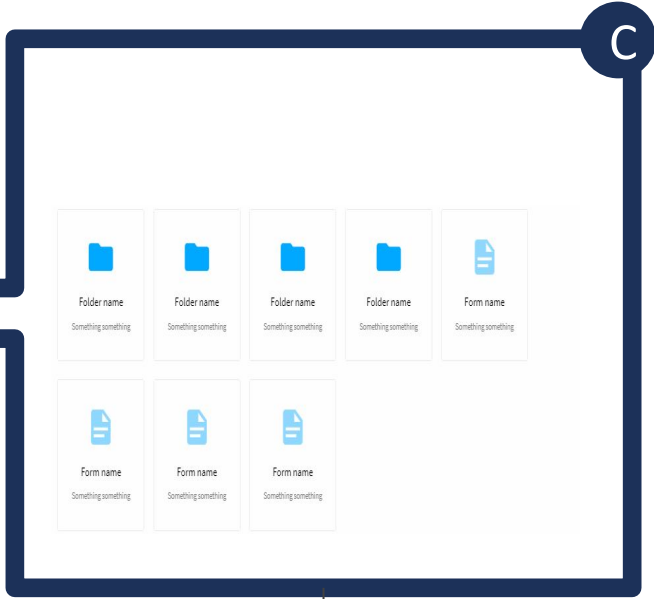
FAIR-DSM – Level 0 Use Case Example

FAIR?...what's FAIR?

We're just happy keeping our data to ourselves, we know it all, no one else needs to know anything about our data.

Data Usage: Single Analysis-driven





R

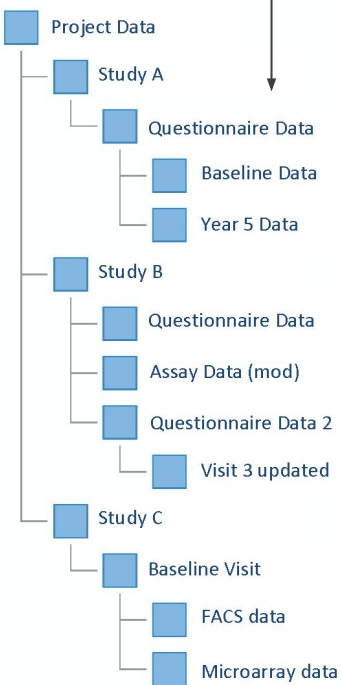
Clinical Chemistry is concerned with quantitative analysis of substances found in the blood or blood serum. The supplementary tables containing **Clinical Haematological Findings**, **Clinical Haemostasis Findings & Urinalysis Findings** being also quantitative follow the same format as this with only the Clinical Chemistry Parameter field replaced by the appropriate parameter name [Clinical Haematology Parameter/ Clinical Haemostasis Parameter/ Urinalysis Parameter]:

Clinical Chemical Findings:	Description:
Average Fold Change	Average fold change (only if explicitly stated in the report)
Average Value	Average value (number only, units captured separately)
Clinical Chemistry Parameter	Clinical chemistry parameter taken
Dose (mg/kg)	Dose in mg/kg
Finding	Finding ie whether increased or decreased (only if explicitly stated in the report)
Relevance	Relevance ie whether considered treatment related or not treatment related (only if explicitly stated in the report)
SD	Standard deviation (not standard error mean)
Sex	Sex (extracted as is from the report)

Page 3 of 8

PERSONAL COMPUTER

PERSONAL COMPUTER



Example of Level 0 Data File

Data files on local storage



Only understood by Data Owner

MAAS-25-9-17
▶ 2017-09-25 [14-27-19-836]
▶ 2017-09-25 [14-30-08-904]
▼ 2017-09-25 [14-34-59-101]
▼ 2017-09-25 [14-34-59-101]
question11_Data.csv
question11_Desc.csv
question112_Data.csv
question112_Desc.csv
question113_Data.csv
question113_Desc.csv
question114_Data.csv
question114_Desc.csv
Recruitment_Data.csv
Recruitment_Desc.csv
Socio8_Data.csv
Socio8_Desc.csv
subjects_Data.csv
subjects_Desc.csv

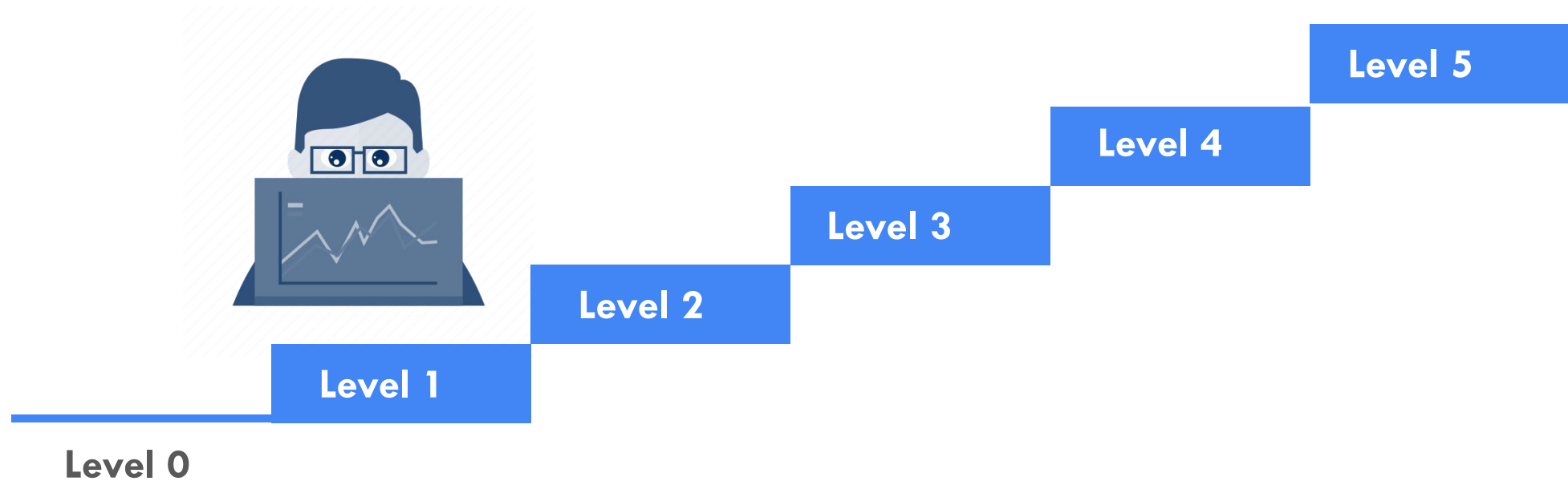


SubjectID	Wheezeever	WheezeXXmths	Wheeze57mths	WheezeattacksXXmths	Wheezeleepdisturb	WheezeumbXXmths	Asthmaever	WheezeexerciseXXmths	WheezenoexerciseXXmths
11003	0	-10	-10	-10	-10	-10	0	-10	-10
11005	0	-10	-10	-10	-10	-10	0	-10	-10
11007	1	1	-10	4	2	0	1	1	1
11008	-9	-9	-9	-9	-9	-9	-9	-9	-9
11011	1	1	-10	3	3	1	1	1	1
11024	0	-10	-10	-10	-10	-10	0	-10	-10
11027	0	-10	-10	-10	-10	-10	0	-10	-10
11028	0	-10	-10	-10	-10	-10	0	-10	-10
11036	0	-10	-10	-10	-10	-10	0	-10	-10
11037	1	1	-10	2	1	0	1	0	0
11038	0	-10	-10	-10	-10	-10	0	-10	-10
11063	0	-10	-10	-10	-10	-10	0	-10	-10
11069	1	1	-10	2	2	0	0	1	1
11074	1	1	-10	2	3	0	1	0	1
11078	0	-10	-10	-10	-10	-10	0	-10	-10
11082	0	-10	-10	-10	-10	-10	0	-10	-10
11083	0	-10	-10	-10	-10	-10	0	-10	-10
11085	0	-10	-10	-10	-10	-10	0	-10	-10
11091	-9	-9	-9	-9	-9	-9	-9	-9	-9
11107	-9	-9	-9	-9	-9	-9	-9	-9	-9
11109	0	-10	-10	-10	-10	-10	0	-10	-10
11110	0	-10	-10	-10	-10	-10	0	-10	-10
11119	0	-10	-10	-10	-10	-10	0	-10	-10
11120	0	-10	-10	-10	-10	-10	0	-10	-10
11125	0	-10	-10	-10	-10	-10	0	-10	-10

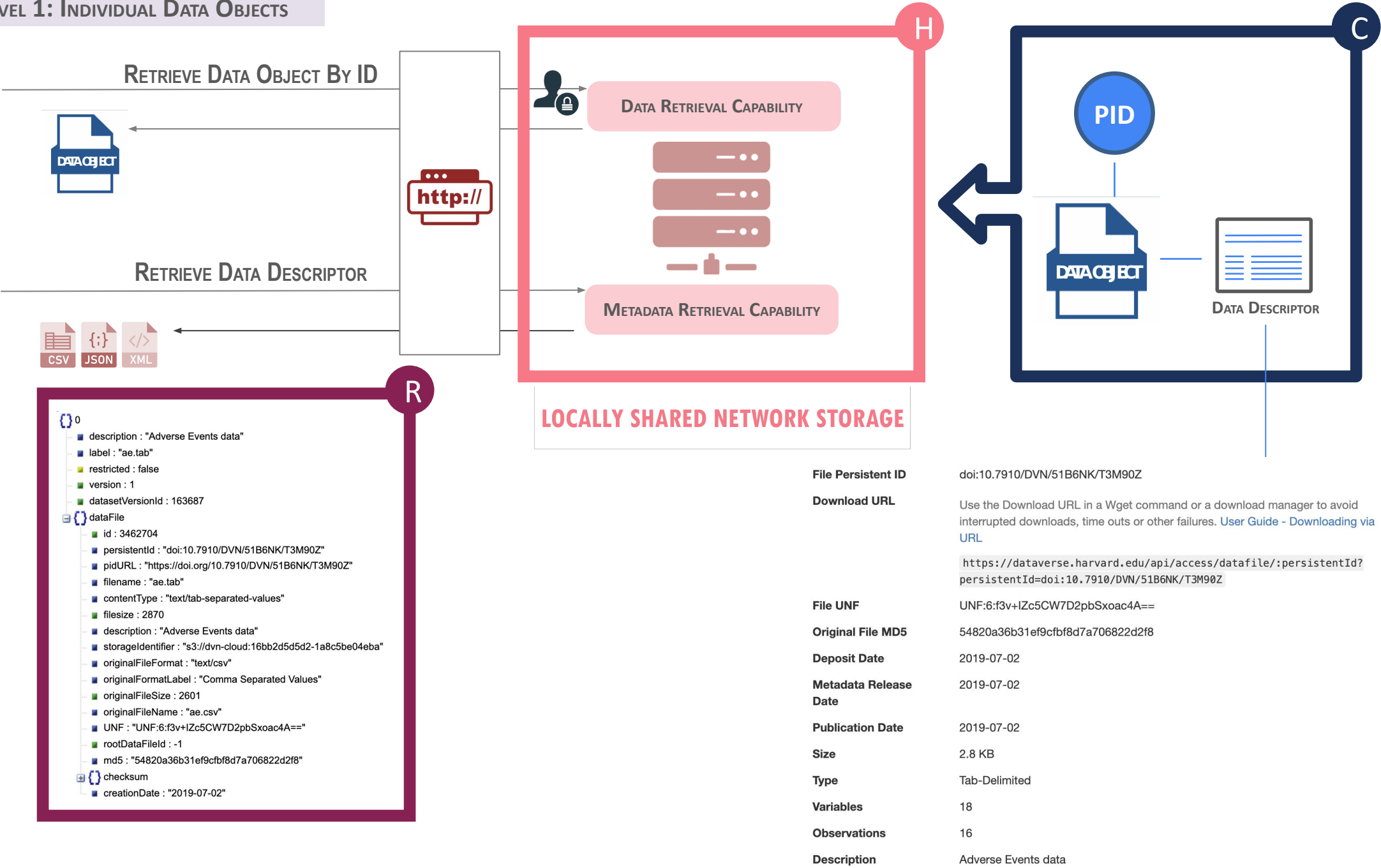
FAIR-DSM – Level 1 Use Case Example

As a researcher I want to publish my analysis-related data, making them Findable, Citable and available for re-use for purpose of future analysis validation

Data Usage Goals: Findability, Sharing, Publish



FAIRPLUS-DSM LEVEL 1: INDIVIDUAL DATA OBJECTS



Level 1 still unmet

MAAS-25-9-17

- 2017-09-25 [14-27-19-836]
- 2017-09-25 [14-30-08-904]
- 2017-09-25 [14-34-59-101]
 - 2017-09-25 [14-34-59-101]
 - question11_Data.csv
 - question11_Desc.csv
 - question112_Data.csv
 - question112_Desc.csv
 - question113_Data.csv
 - question113_Desc.csv
 - question114_Data.csv
 - question114_Desc.csv
 - Recruitment_Data.csv
 - Recruitment_Desc.csv
 - Socio8_Data.csv
 - Socio8_Desc.csv
 - subjects_Data.csv
 - subjects_Desc.csv



DATA FILE									
SubjectID	Wheezeever	WheezeXmths	Wheeze57mths	WheezeattacksXmths	Wheezeleepdisturb	WheezedumbXmths	Asthmaever	WheezeexerciseXmths	WheezenoexerciseXmths
11003	0	-10	-10	-10	-10	-10	0	-10	-10
11005	0	-10	-10	-10	-10	-10	0	-10	-10
11007	1	1	-10	4	2	0	1	1	1
11008	-9	-9	-9	-9	-9	-9	-9	-9	-9
11011	1	1	-10	3	3	1	1	1	1
11024	0	-10	-10	-10	-10	-10	0	-10	-10
11027	0	-10	-10	-10	-10	-10	0	-10	-10
11028	0	-10	-10	-10	-10	-10	0	-10	-10
11036	0	-10	-10	-10	-10	-10	0	-10	-10
11037	1	1	-10	2	1	0	1	0	0
11038	0	-10	-10	-10	-10	-10	0	-10	-10
11063	0	-10	-10	-10	-10	-10	0	-10	-10
11069	1	1	-10	2	2	0	0	1	1
11074	1	1	-10	2	3	0	1	0	1
11078	0	-10	-10	-10	-10	-10	0	-10	-10
11082	0	-10	-10	-10	-10	-10	0	-10	-10
11083	0	-10	-10	-10	-10	-10	0	-10	-10
11085	0	-10	-10	-10	-10	-10	0	-10	-10
11091	-9	-9	-9	-9	-9	-9	-9	-9	-9
11107	-9	-9	-9	-9	-9	-9	-9	-9	-9
11109	0	-10	-10	-10	-10	-10	0	-10	-10
11110	0	-10	-10	-10	-10	-10	0	-10	-10
11119	0	-10	-10	-10	-10	-10	0	-10	-10
11120	0	-10	-10	-10	-10	-10	0	-10	-10
11125	0	-10	-10	-10	-10	-10	0	-10	-10

DESCRIPTOR FILE				
TemplateVersion	"question113"	1		
DatasetIdentifier	"MAAS: 11 Year Environmental and Respiratory Questionnaire"			
DatasetDescription	"Questionnaire" "Respiratory Symptoms" "Age 11"			
DatasetTags	Rule1."question113";"SubjectID";"respiratory11yr";"SubjectID"			
rule	Rule2."question113";"SubjectID";"AIW";"SubjectID"			
Resources	<Quest11>			
NsTemplateDefault	stelar/catalogue/template/			
NsParameterDefault	stelar/catalogue/Parameter/			
NsTermDefault	stelar/catalogue/Term/			
Alias	ColumnIdentifier	DropOrNot	ColumnDescription	ColumnTags
"SubjectID"	"SubjectID"	0		
"Wheezeever"	"Wheezeever"	0	QuestionResponse(Dat "Age 11 FollowUp" "wheeze" "whistle" "asthma" "Ever" "CrumbW"	
"WheezeXmths"	"WheezeXmths"	0	QuestionResponse(Dat "Age 11 FollowUp" "wheeze" "whistle" "asthma"	
"Wheeze57mths"	"Wheeze57mths"	0	QuestionResponse(Dat "Age 11 FollowUp" "wheeze" "whistle" "asthma" "Current" "CrumbW"	
"WheezeattacksXmths"	"WheezeattacksXmths"	0	QuestionResponse(Dat "Age 11 FollowUp" "wheeze" "whistle" "asthma" "Current" "CrumbW"	
"Wheezeleepdisturb"	"Wheezeleepdisturb"	0	QuestionResponse(Dat "Age 11 FollowUp" "wheeze" "whistle" "asthma" "Current" "CrumbW"	
"WheezedumbXmths"	"WheezedumbXmths"	0	QuestionResponse(Dat "Age 11 FollowUp" "wheeze" "whistle" "asthma"	
"Asthmaever"	"Asthmaever"	0	QuestionResponse(Dat "Age 11 FollowUp" "asthma" "doctor diagnosis" "Ever" "CrumbA"	
"WheezeexerciseXmths"	"WheezeexerciseXmths"	0	QuestionResponse(Dat "Age 11 FollowUp" "wheeze" "whistle" "asthma" "Current" "CrumbW"	
"WheezenoexerciseXmths"	"WheezenoexerciseXmths"	0	QuestionResponse(Dat "Age 11 FollowUp" "wheeze" "whistle" "asthma" "Ever" "CrumbW"	
"WheezefluXmths"	"WheezefluXmths"	0	QuestionResponse(Dat "Age 11 FollowUp" "wheeze" "whistle" "asthma" "Ever" "CrumbW"	
"WheezenofluXmths"	"WheezenofluXmths"	0	QuestionResponse(Dat "Age 11 FollowUp" "wheeze" "whistle" "asthma"	
"Wheezeobxmxmths"	"Wheezeobxmxmths"	0	QuestionResponse(Dat "Age 11 FollowUp" "sleep" "shortness of breath"	
"WheezechestXmths"	"WheezechestXmths"	0	QuestionResponse(Dat "Age 11 FollowUp" "sleep" "chest tightness"	
"WheezeworseXmths"	"WheezeworseXmths"	0	QuestionResponse(Dat "Age 11 FollowUp" "wheeze" "trigger" "exacerbation"	

Content [DSM-1-C0] Each Dataset purposed for FAIR sharing and reuse is assigned a unique identifier

Hosting Env. Capabilities [DSM-1-H2] The Dataset and its Descriptor are indexed and retrievable (in the same or separate hosting environments) via unique and persistent identifiers

Example of Level 1 Data

A generically described data file hosted by Harvard Dataverse



Other Examples of general-purpose data hosting environments:



ae.tab

This file is part of "CDISC-SDTM sample study".

Version 1.1

File Citation

Emam, Ibrahim; CDISC, 2019, "ae.tab", *CDISC-SDTM sample study*, <https://doi.org/10.7910/DVN/51B6NK/T3M90Z>, Harvard Dataverse, V1, UNF:6:f3v+IZc5CW7D2pbSxoac4A== [fileUNF]

[Cite Data File](#) [Learn about Data Citation Standards.](#)

Dataset Citation

Emam, Ibrahim; CDISC, 2019, "CDISC-SDTM sample study", <https://doi.org/10.7910/DVN/51B6NK>, Harvard Dataverse, V1, UNF:6:TlJzJErXxQzFcLkrYJulg== [fileUNF]

[Cite Dataset](#) [Learn about Data Citation Standards.](#)

[Preview](#) [Metadata](#) [Versions](#)

[File Metadata](#)

Export Metadata

- Dublin Core
- DDI
- DataCite
- DDI HTML Codebook
- JSON
- OAI_ORE
- OpenAIRE
- Schema.org JSON-LD

Preview



File Persistent ID doi:10.7910/DVN/51B6NK/T3M90Z

Download URL

Use the Download URL in a Wget command or a download manager to avoid interrupted downloads, time outs or other failures.
[User Guide - Downloading via URL](#)

<https://dataverse.harvard.edu/api/access/datafile/:persistentId?persistentId=doi:10.7910/DVN/51B6NK/T3M90Z>

File UNF UNF:6:f3v+IZc5CW7D2pbSxoac4A==

Original File MD5 54820a36b31ef9cfb8d7a706822d2f8

Deposit Date 2019-07-02

Metadata Release Date 2019-07-02

Publication Date 2019-07-02

Size 2.8 KB

Type Tab-Delimited

Variables 18

Observations 16

Description Adverse Events data

HARVARD
Dataverse

Add Data ▾ Search ▾ About User Guide Support Sign Up Log In

Harvard Dataverse >

CDISC-SDTM sample study

Version 1.1

Emam, Ibrahim; CDISC, 2019, "CDISC-SDTM sample study", <https://doi.org/10.7910/DVN/51B6NK>, Harvard Dataverse, V1, UNF:6:TjZjEreXxQzFcLkYJulg== [fileUNF]

Cite Dataset ▾ Learn about Data Citation Standards.

Access Dataset ▾
Contact Owner Share

Dataset Metrics
6,655 Downloads

Description
This is a sample study dataset containing CDISC-SDTM formatted data files created originally by CDISC for demo purposes. This dataset can be used by anyone who is interested in CDISC-SDTM formatted dataset. This version is used to demonstrate loading standard clinical datasets into PlatformTM: a data custodianship platform for translational research data. Live demo here: <https://platformtm.cloud/app/#/projects/15/main>

Subject
Medicine, Health and Life Sciences

Keyword
Clinical data

License/Data Use Agreement
CC0 1.0

Files **Metadata** Terms Versions

Search this dataset...

Filter by
File Type: All ▾ Access: All ▾

1 to 7 of 7 Files

	ae.tab Tabular Data - 2.8 KB Published Jul 2, 2019 1,517 Downloads 18 Variables, 16 Observations Adverse Events data	 
	dm.tab Tabular Data - 862 B Published Jul 2, 2019 1,154 Downloads 16 Variables, 5 Observations Subject demographics data	 
	eg.tab Tabular Data - 9.0 KB Published Jul 2, 2019 757 Downloads 21 Variables, 56 Observations Electrocardiogram data	 
	lb.tab Tabular Data - 16.5 KB Published Jul 2, 2019 848 Downloads	 

Files grouped together in a dataset

Files **Metadata** Terms Versions

Export Metadata ▾

Dataset DOI

Citation Metadata ▾

Dataset Persistent ID	doi:10.7910/DVN/51B6NK
Publication Date	2019-07-02
Title	CDISC-SDTM sample study
Author	Emam, Ibrahim (Imperial College London) CDISC
Contact	Use email button above to contact. Emam, Ibrahim (Imperial College London)
Description	This is a sample study dataset containing CDISC-SDTM formatted data files created originally by CDISC for demo purposes. This dataset can be used by anyone who is interested in CDISC-SDTM formatted dataset. This version is used to demonstrate loading standard clinical datasets into PlatformTM: a data custodianship platform for translational research data. Live demo here: https://platformtm.cloud/app/#/projects/15/main
Subject	Medicine, Health and Life Sciences
Keyword	Clinical data
Related Publication	https://platformtm.cloud/app/#/projects/15/main
Depositor	Emam, Ibrahim
Deposit Date	2019-07-02

Study/project-level metadata

ae.tab
This file is part of "CDISC-SDTM sample study".

Version 1.1

File Citation
Emam, Ibrahim; CDISC, 2019, "ae.tab", CDISC-SDTM sample study, <https://doi.org/10.7910/DVN/51B6NK/T3M90Z>, Harvard Dataverse, V1, UNF:6:3dv+Izc5CW7D2pbSxoac4A== [fileUNF]

Cite Data File ▾ Learn about Data Citation Standards.

Dataset Citation
Emam, Ibrahim; CDISC, 2019, "CDISC-SDTM sample study", <https://doi.org/10.7910/DVN/51B6NK>, Harvard Dataverse, V1, UNF:6:TjZjEreXxQzFcLkYJulg== [fileUNF]

Cite Dataset ▾ Learn about Data Citation Standards.

Previous **Metadata** Versions

Export Metadata ▾

File identifier

File Metadata ▾

Preview



File Persistent ID
doi:10.7910/DVN/51B6NK/T3M90Z

Download URL
Use the Download URL in a Wget command or a download manager to avoid interrupted downloads, time outs or other failures.
User Guide - Downloading via URL
<https://dataverse.harvard.edu/api/access/datafile/persistentId?persistentId=doi:10.7910/DVN/51B6NK/T3M90Z>

File UNF
Original File MD5
54820a38b31ef9c8b8d7a706822d2f8

Deposit Date
2019-07-02

Metadata Release Date
2019-07-02

Publication Date
2019-07-02

Size
2.8 KB

Type
Tab-Delimited

Variables
18

Observations
16

File metadata

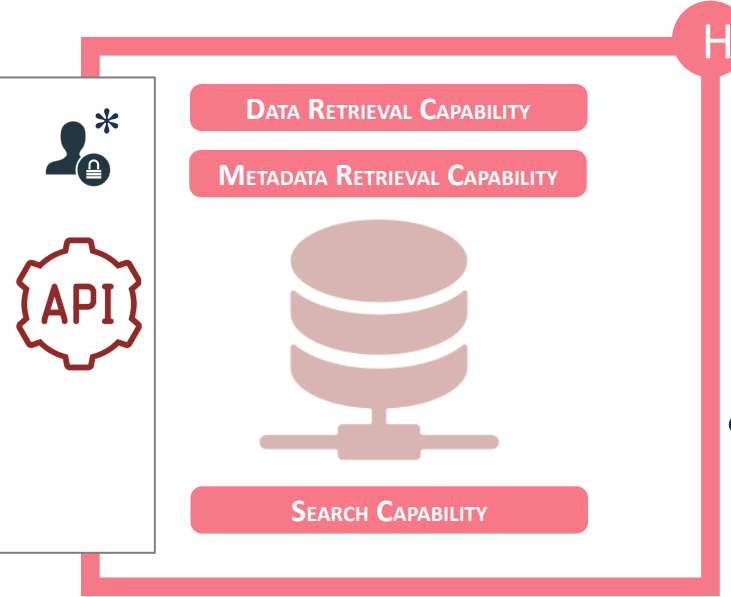
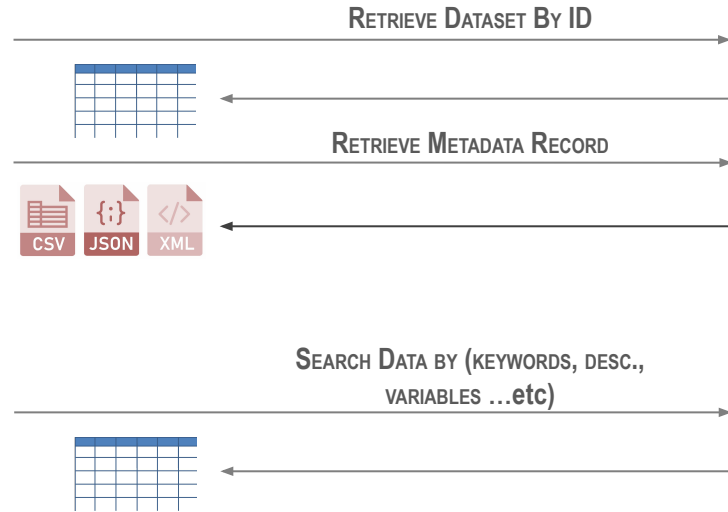
FAIR-DSM – Level 2 Use Case

As a Project Data Manager/Steward/Custodian, I want to standardise all project data outputs maintaining the relationship between the various data modalities within the context of the project, to preserve the data and support multiple analyses and multiple researchers in my project

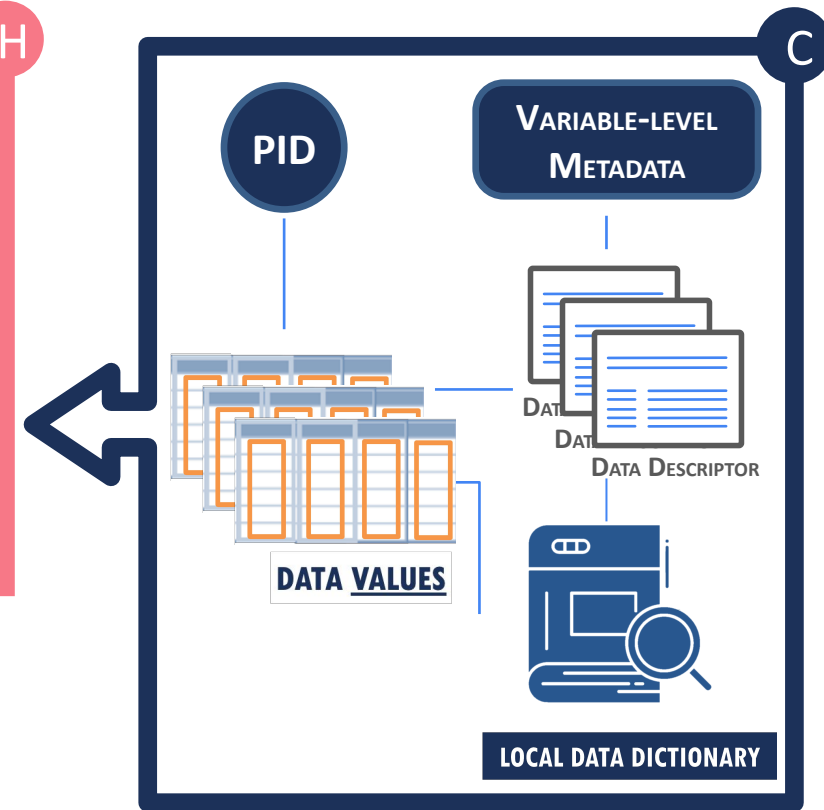
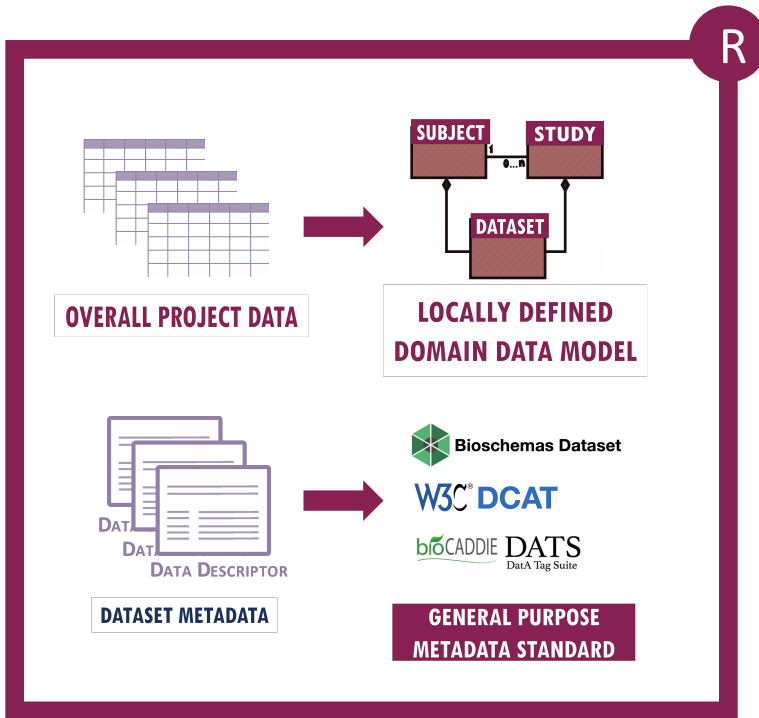
Data Usage Goals: Data Exploration, Data Visualisation, Re-use: Support for multiple analyses within project



FAIRPLUS-DSM LEVEL 2 : PROJECT / INSTITUTIONAL LEVEL



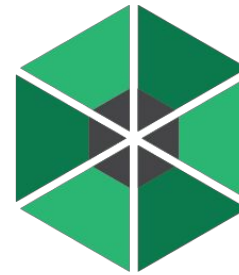
PROJECT-SPECIFIC
OR
INSTITUTIONAL DATA REPOSITORY



Example of Level 2 Data

Data values are uniformly mapped to a locally defined dictionary of terms

ColumnIdentifier	ValueLabels
"Wheezesleepdisturb"	1="Never Woken With Wheeze" 2="Less Than One Night Per Week" 3="More Than One Night Per Week"
"Wheezedumbxxmths"	^-9="Subject Did Not Attend" -10="Skipped Question"
"Asthmaever"	0="No" 1="Yes" 999="Don't Know" -9="Subject Did Not Attend" -10="Skipped Question"
"WheezeeexerciseXXmths"	0="No" 1="Yes" 999="Don't Know" -9="Subject Did Not Attend" -10="Skipped Question"
"WheezenoexerciseXXmths"	0="No" 1="Yes" 999="Don't Know" -9="Subject Did Not Attend" -10="Skipped Question"
"WheezefluXXmths"	0="No" 1="Yes" 999="Don't Know" -9="Subject Did Not Attend" -10="Skipped Question"
"WheezenofluXXmths"	0="No" 1="Yes" 999="Don't Know" -9="Subject Did Not Attend" -10="Skipped Question"
"Wheezesobxxmths"	0="No" 1="Yes" 999="Don't Know" -9="Subject Did Not Attend" -10="Skipped Question"
"WheezetightchestXXmths"	0="No" 1="Yes" 999="Don't Know" -9="Subject Did Not Attend" -10="Skipped Question"
"WheezeworseXXmths"	^-9="Subject Did Not Attend" -10="Skipped Question"
"Wheezeothers1"	^-9="Subject Did Not Attend" -10="Skipped Question"
"Wheezeothers2"	^-9="Subject Did Not Attend" -10="Skipped Question"
"Wheezeothers3"	^-9="Subject Did Not Attend" -10="Skipped Question"
"Wheezeothers4"	^-9="Subject Did Not Attend" -10="Skipped Question"
"Wheezeothers5"	^-9="Subject Did Not Attend" -10="Skipped Question"
"Wheezeothers6"	^-9="Subject Did Not Attend" -10="Skipped Question"
"Asthmamedication"	0="No" 1="Yes" 999="Don't Know" -9="Subject Did Not Attend" -10="Skipped Question"
"Asthmapmed1freq"	"Regular"="Regularly" "Irritated"="When Irritated" "RegularIrritated"="Both when Irritated & Regularly"
"Asthmapmed2freq"	"Regular"="Regularly" "Irritated"="When Irritated" "RegularIrritated"="Both when Irritated & Regularly"
"Asthmapmed3freq"	"Regular"="Regularly" "Irritated"="When Irritated" "RegularIrritated"="Both when Irritated & Regularly"
"Asthmapmed4freq"	"Regular"="Regularly" "Irritated"="When Irritated" "RegularIrritated"="Both when Irritated & Regularly"
"Asthmapmed5freq"	"Regular"="Regularly" "Irritated"="When Irritated" "RegularIrritated"="Both when Irritated & Regularly"
"Asthmacmed1freq"	"Regular"="Regularly" "Irritated"="When Irritated" "RegularIrritated"="Both when Irritated & Regularly"
"Asthmacmed2freq"	"Regular"="Regularly" "Irritated"="When Irritated" "RegularIrritated"="Both when Irritated & Regularly"
"Asthmapmed1"	^-9="Subject Did Not Attend" -10="Skipped Question"
"Asthmapmed2"	^-9="Subject Did Not Attend" -10="Skipped Question"
"Asthmapmed3"	^-9="Subject Did Not Attend" -10="Skipped Question"
"Asthmapmed4"	^-9="Subject Did Not Attend" -10="Skipped Question"
"Asthmapmed5"	^-9="Subject Did Not Attend" -10="Skipped Question"
"Asthmacmed1"	^-9="Subject Did Not Attend" -10="Skipped Question"
"Asthmacmed2"	^-9="Subject Did Not Attend" -10="Skipped Question"
"Asthmamedicationex"	0="No" 1="Yes" 999="Don't Know" -9="Subject Did Not Attend" -10="Skipped Question"
"Asthmapmed1freqex"	"Regular"="Regularly" "Irritated"="When Irritated" "RegularIrritated"="Both when Irritated & Regularly"
"Asthmapmed2freqex"	"Regular"="Regularly" "Irritated"="When Irritated" "RegularIrritated"="Both when Irritated & Regularly"
"Asthmapmed3freqex"	"Regular"="Regularly" "Irritated"="When Irritated" "RegularIrritated"="Both when Irritated & Regularly"



Bioschemas

```
{
  "@type": "Dataset",
  "name": "MAAS RESPY5",
  "description": "MAAS Respiratory Questionnaire Data Year 5",
  "keywords": ["Asthma", "Respiratory", "Cohort Study"],
  "measurementTechnique": "Clinical Questionnaire",
  "url": "http://xyz.ds.ac.uk/unicorn/MTBLS234",
  "license": "http://xyz.ds.ac.uk/unicorn/terms-of-use",
  "variableMeasured": [
    { "@type": "schema:PropertyValue",
      "name": "Wheeze sleep disturbance" },
    { "@type": "schema:PropertyValue",
      "name": "Asthma Medications" },
    { "@type": "schema:PropertyValue",
      "name": "Asthma Medication Frequency" },
    { "@type": "schema:PropertyValue",
      "name": "Chest Tightness" }
  ]
}
```

Example of Level 2 Data

Content	[DSM-2-C2]	Where applicable, the Dataset Model organises data values within a dataset according to the Tidy Data Principles
Content	[DSM-2-C6]	Dataset Descriptor includes Field-level Metadata as prescribed by the adopted Dataset Model



is dependent on

Two variables per field: Visit & Observed Feature



SubjId	Cohort	sysbp_sc	diabp_sc	heart_rate_sc	sysbp_fu	diabp_fu	heart_rate_fu
s-01	cohort_A	100	65	89	95	63	70
s-02	cohort_A	134	77	60	120	71	66
s-03	cohort_B	182	122	110	190	100	100
s-04	cohort_B	112	81	90	120	98	78
s-05	cohort_a	127	74	70	135	75	72

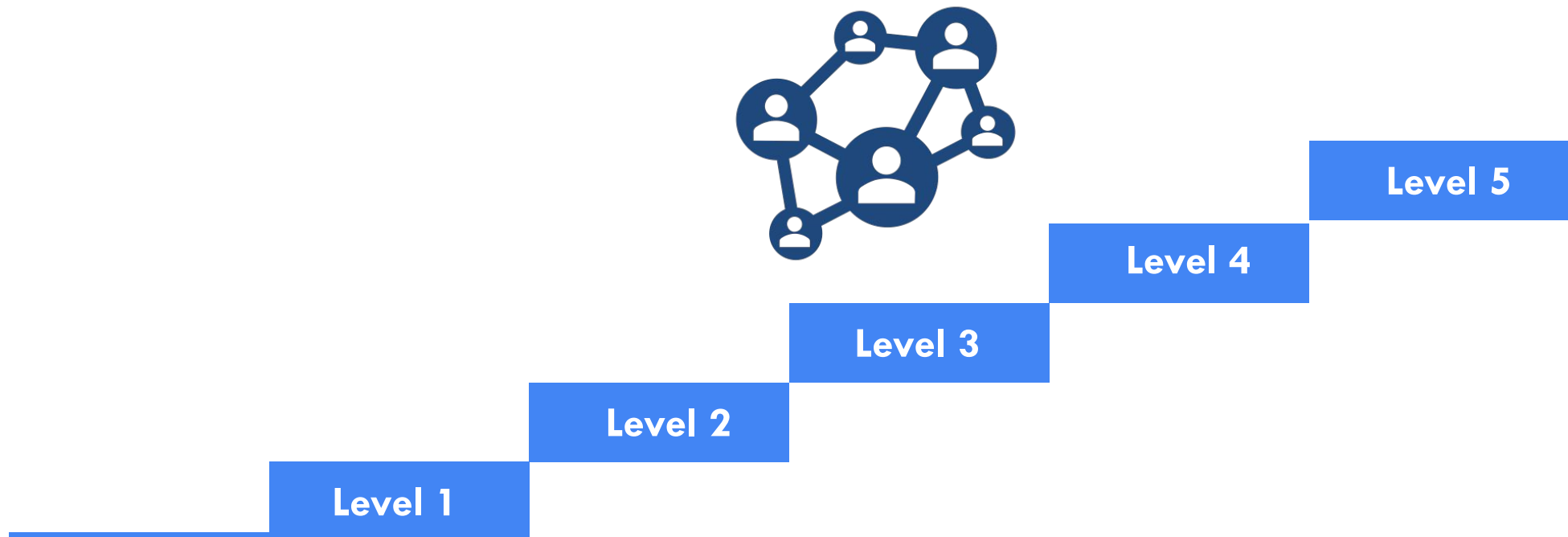


SubjId	Cohort	sysbp	diabp	heart_rate	Visit
s-01	cohort_A	100	65	89	Screening
s-02	cohort_A	134	77	60	Screening
s-03	cohort_B	182	122	110	Screening
s-04	cohort_B	112	81	90	Screening
s-01	cohort_A	95	63	70	Follow-up
s-02	cohort_A	120	71	66	Follow-up
s-03	cohort_B	190	100	100	Follow-up
s-04	cohort_B	120	98	78	Follow-up

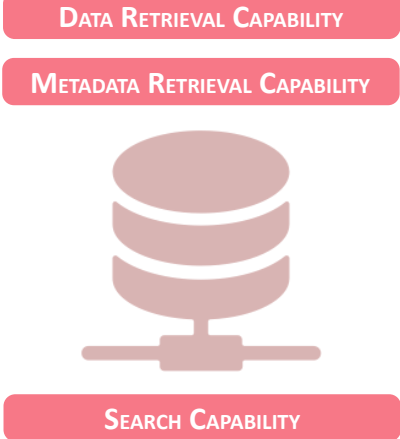
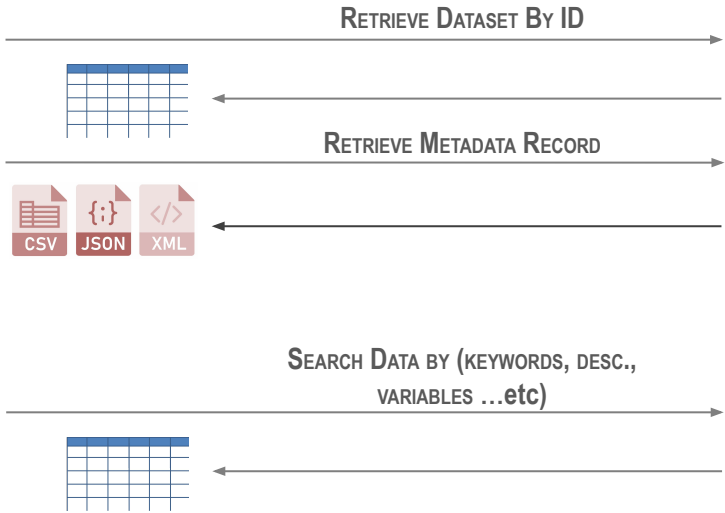
FAIR-DSM – Level 3 Use Case

As a project consortium, we want to guarantee long term preservation of our data and to be shared with the community; to enable re-purposing of our data and the ability to integrate with other data within the same domain / subject area.

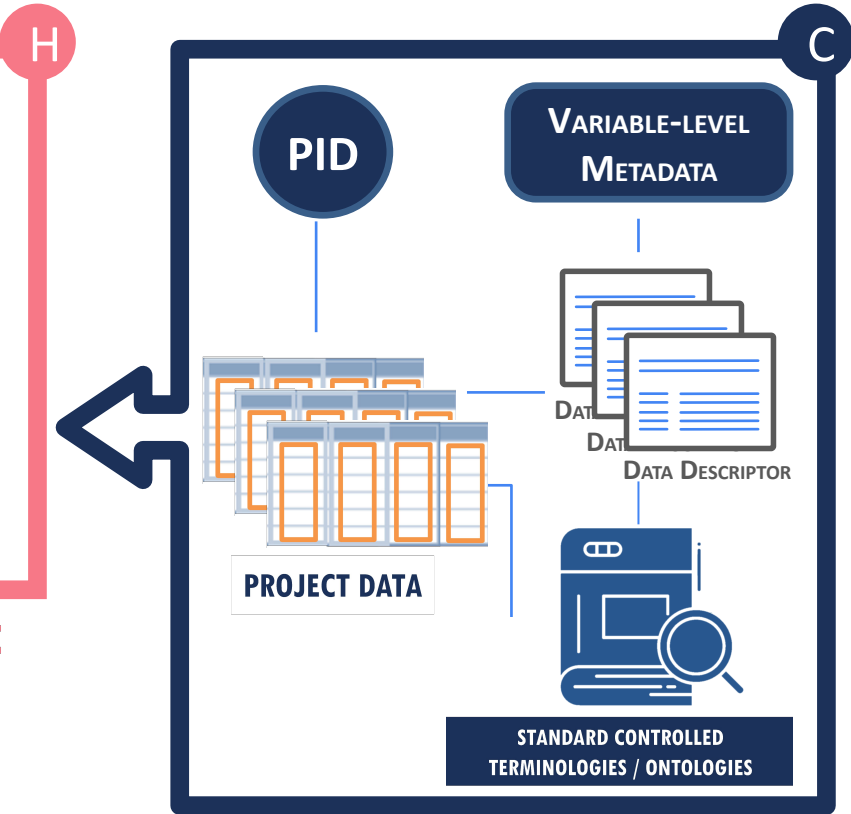
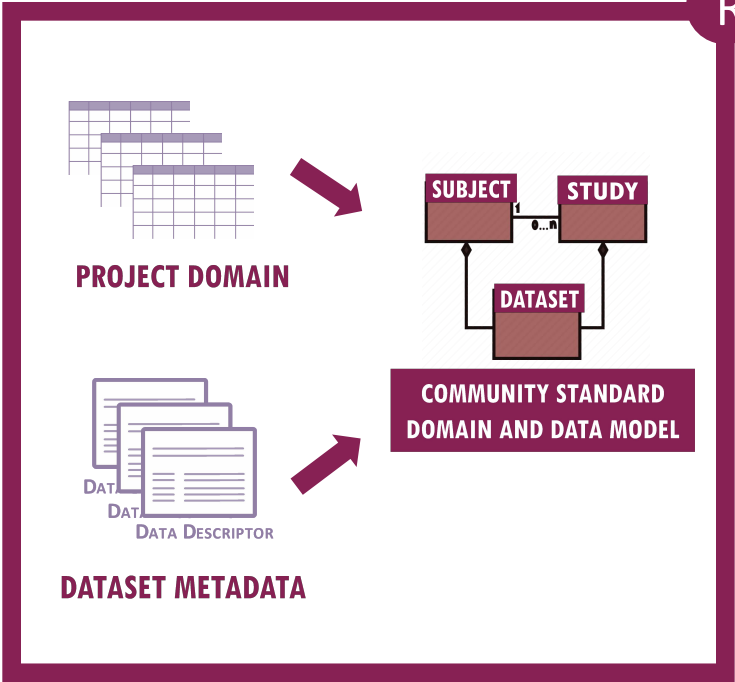
Data Usage Goals: Data Usage Goals: Data Integration, Data Preservation, Re-use



FAIRPLUS-DSM LEVEL 3 : COMMUNITY LEVEL



COMMUNITY / DOMAIN SPECIFIC
DATA REPOSITORY

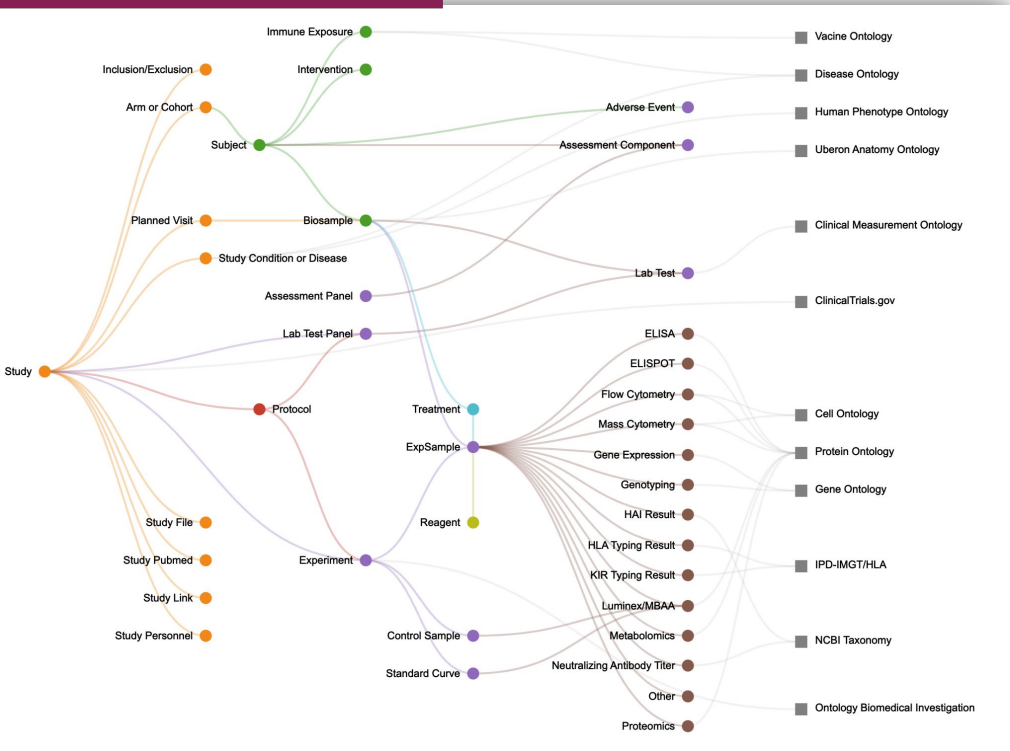


Example of Level 3 Data

Study Data hosted in community-based / domain-specific data repository



PROJECT DOMAIN DATA MODEL



SDY283 - Exploration of molecular mechanism(s) for TLR ligand response differences between adults and neonates

[Download](#)

[Summary](#) [Design](#) [Adverse Event](#) [Assessment](#) [Interventions](#) [Medications](#) [Substance](#) [Demographics](#) [Lab Tests](#) [Mechanistic Assays](#) [Study Files](#)

+ -

Summary

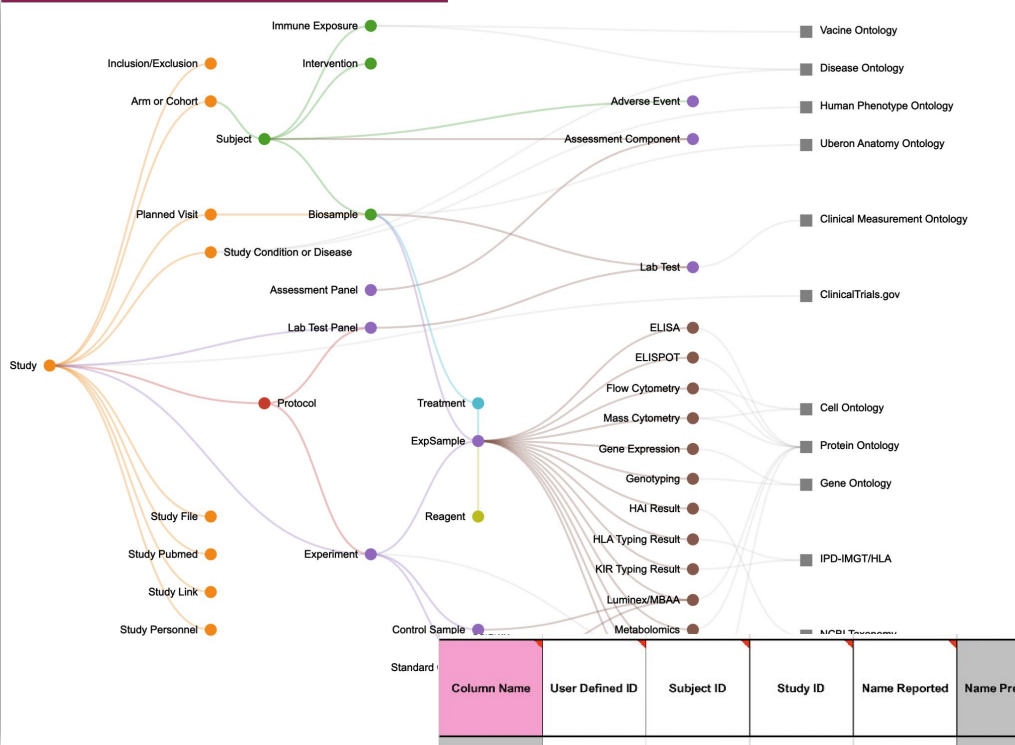
Accession	SDY283
Title	Exploration of molecular mechanism(s) for TLR ligand response differences between adults and neonates
DOI	10.21430/M30USX09NF
Brief Description	The study will perform expression microarrays on RNA from purified subsets of neonatal and adult APCs either stimulated or not stimulated with TLR ligands of interest. Arrays will be performed on cells from neonatal and adult individuals and the arrays for all will be run in parallel on the same day.
PI	Christopher Wilson - University of Washington
Type	
Condition Studied	Aging
Start Date	
Detailed Description	The study performed expression microarrays on RNA from purified subsets of neonatal and adult APCs, which were stimulated or not stimulated with a TLR ligands of interest. Arrays were performed on cells from neonatal and adult individuals, and the arrays for all will be run in parallel on the same day. This study used ELISA to assess the abundance and activation induced modifications of relevant TLRs, adaptors and downstream signaling pathway components.
Objectives	To explore the molecular mechanism(s) for TLR ligand response differences between adults and neonates and to assess abundance and activation induced modifications of relevant TLRs, adaptors and downstream signaling pathway components.
Hypothesis	none
Endpoints	ELISA
Gender Included	Not Specified
Subjects Number	89
Download Packages1	Study Download Packages
Contract/Grant	TLRs in Innate Immunity and the Induction of Adaptive Immunity in the Neonate and Infant
Program	Immune Function and Biodefense in Children, Elderly, and Immunocompromised Populations
Data Completeness	2 - Complete set of descriptive data and results, as ascertained by ImmPort.

Example of Level 3 Data

Study Data hosted in community-based / domain-specific data repository



PROJECT DOMAIN DATA MODEL



SDY283 - Exploration of molecular mechanism(s) for TLR ligand response differences between adults and neonates

[Download](#)

Summary Design **Adverse Event** Assessment Interventions Medications Substance Demographics Lab Tests Mechanistic Assays Study Files

+ -

Adverse Event Summary

Search Adverse Event Summary



Export

Totals By	ARM1910	ARM1911
Grade 1 Mild Adverse Event	0	0
Grade 2 Moderate Adverse Event	0	0
Grade 3 Severe Adverse Event	0	0
Grade 4 Life Threatening or Disabling Adverse Event	0	0
Grade 5 Death Related to Adverse Event	0	0
Not Specified	0	0
Subject	64	25
Subject with Adverse Event	0	0
Total Adverse Event	0	0

<< < 1 > >> 40

Showing 1 to 9 of 9



DATASET METADATA / DATA MODEL

Column Name	User Defined ID	Subject ID	Study ID	Name Reported	Name Preferred	Severity Reported	Outcome Reported	Start Study Day	End Study Day	Relation To Study Treatment	Organ Or Body System Reported	Description	Location Of Reaction Reported	Study Treatment Action Taken	Relation To Nonstudy Treatment	Causality	Start Time	End Time
-------------	-----------------	------------	----------	---------------	----------------	-------------------	------------------	-----------------	---------------	-----------------------------	-------------------------------	-------------	-------------------------------	------------------------------	--------------------------------	-----------	------------	----------

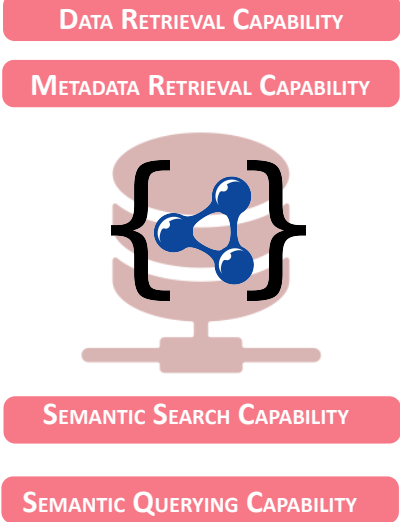
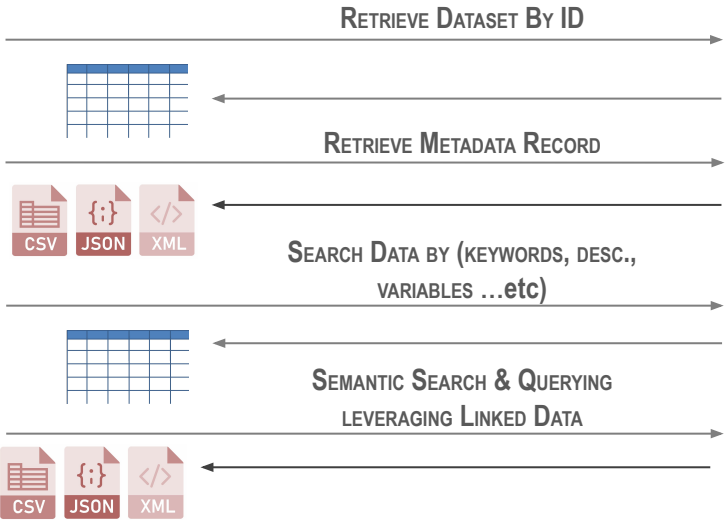
FAIR-DSM – Level 4 Use Case

As a research institute, I want to build a FAIR Data Commons to be able to integrate and harmonise data of interest from different studies and different data resources for the purpose of using data in meta-analyses studies and cross-study analyses.

Data Usage Goals: Meta-analyses, Data Interoperability, Cross-study/domain Data Integration and Harmonisation



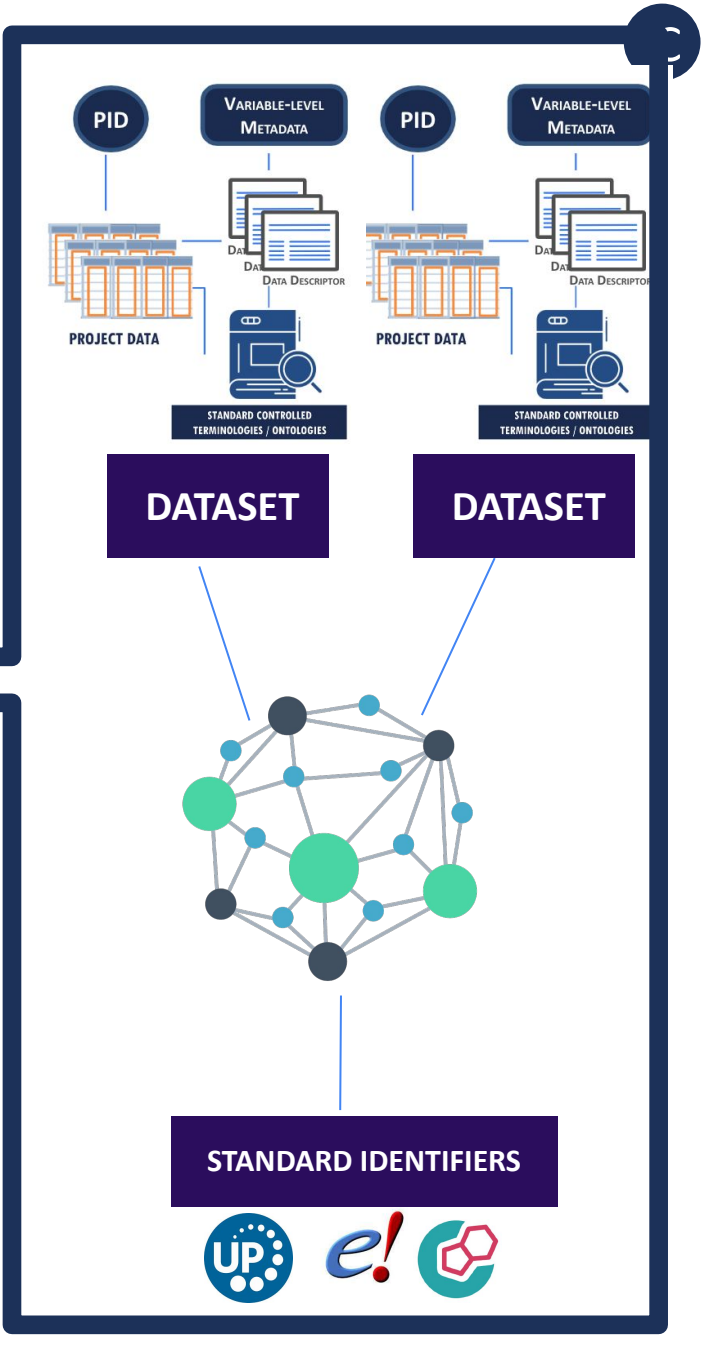
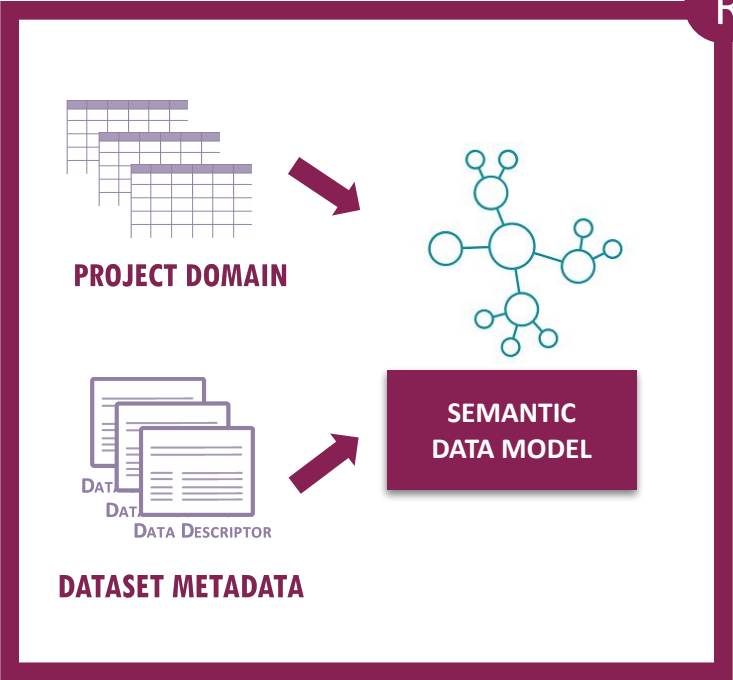
FAIRPLUS-DSM LEVEL 4 : CROSS-DOMAIN/COMMUNITY LEVEL



H

LINKED DATA STORE

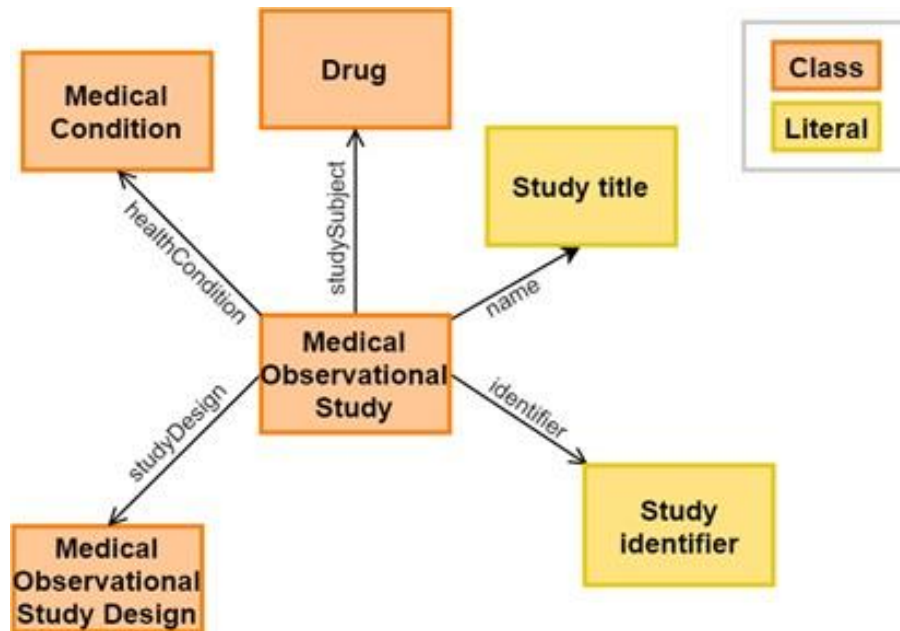
R



Example of Level 4 Data

<https://w3id.org/faircookbook/FCB054>

<https://covid19.ohdsi.app/>



```
schema:MedicalObservationalStudy a owl:Class ;
  rdfs:isDefinedBy
    <https://health-lifesci.schema.org/MedicalObservationalStudy> ;
  rdfs:label "MedicalObservationalStudy"@en ;
  rdfs:subClassOf schema:MedicalStudy .
```

```
schema:Drug a owl:Class ;
  rdfs:isDefinedBy
    <https://health-lifesci.schema.org/Drug> ;
  rdfs:label "Drug"@en ;
  rdfs:subClassOf schema:Substance .
```

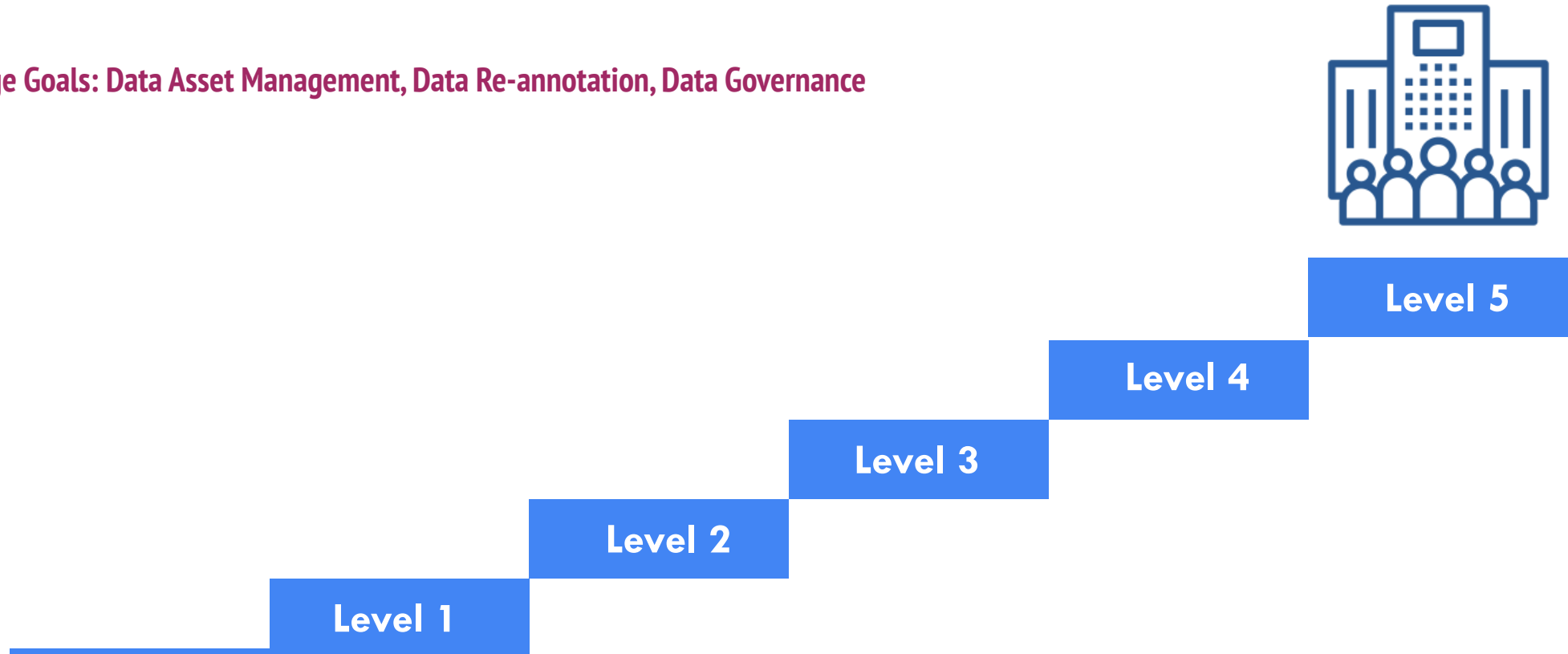
```
{
  "@id": "study:Covid19EstimationHydroxychloroquine",
  "@type": "schema:MedicalObservationalStudy",
  "identifier": "EUPAS34497",
  "name": "OHDSI COVID-19 Studyathon: Hydroxychloroquine population-level ef",
  "description": "Many existing drugs are being considered for use in treat",
  "studyType": ["ohdsi:ClinicalApplication"],
  "analyticsUseCase": ["ohdsi:PopulationLevelEstimation"],
  "mainEntityOfPage": "http://www.encepp.eu/encepp/viewResource.htm?id=34498",
  "studyDesign": "Covid19EstimationHydroxychloroquine/design",
  "status": ["ohdsi:ResultsAvailable"],
  "healthCondition": [
    "concept:4329847",
    "concept:199874",
    "concept:197320",
    "concept:432961",
    "concept:444044",
    "concept:432791",
    "concept:4169095",
    "concept:316999",
    "concept:4171683",
    "concept:4173000"
  ],
  "drug": [
    {
      "@id": "concept:19008260.0",
      "@type": "schema:Drug",
      "code": {
        "@type": "schema:MedicalCode",
        "codeValue": "11196",
        "codingSystem": "RxNorm"
      },
      "identifier": "19008260.0",
      "name": "Viloxazine"
    },
    {
      "@id": "concept:44507700.0",
      "@type": "schema:Drug",
      "code": {
        "@type": "schema:MedicalCode",
        "codeValue": "1455099",
        "codingSystem": "RxNorm"
      },
      "identifier": "44507700.0",
      "name": "vortioxetine"
    },
    {
      "@id": "concept:4329847",
      "@type": "schema:MedicalCondition",
      "code": {
        "@type": "schema:MedicalCode",
        "codeValue": "22298006",
        "codingSystem": "SNOMED",
        "sameAs": "http://snomed.info/id/22298006"
      },
      "identifier": "4329847",
      "name": "Myocardial infarction"
    }
  ]
}
```

Standard
identifiers

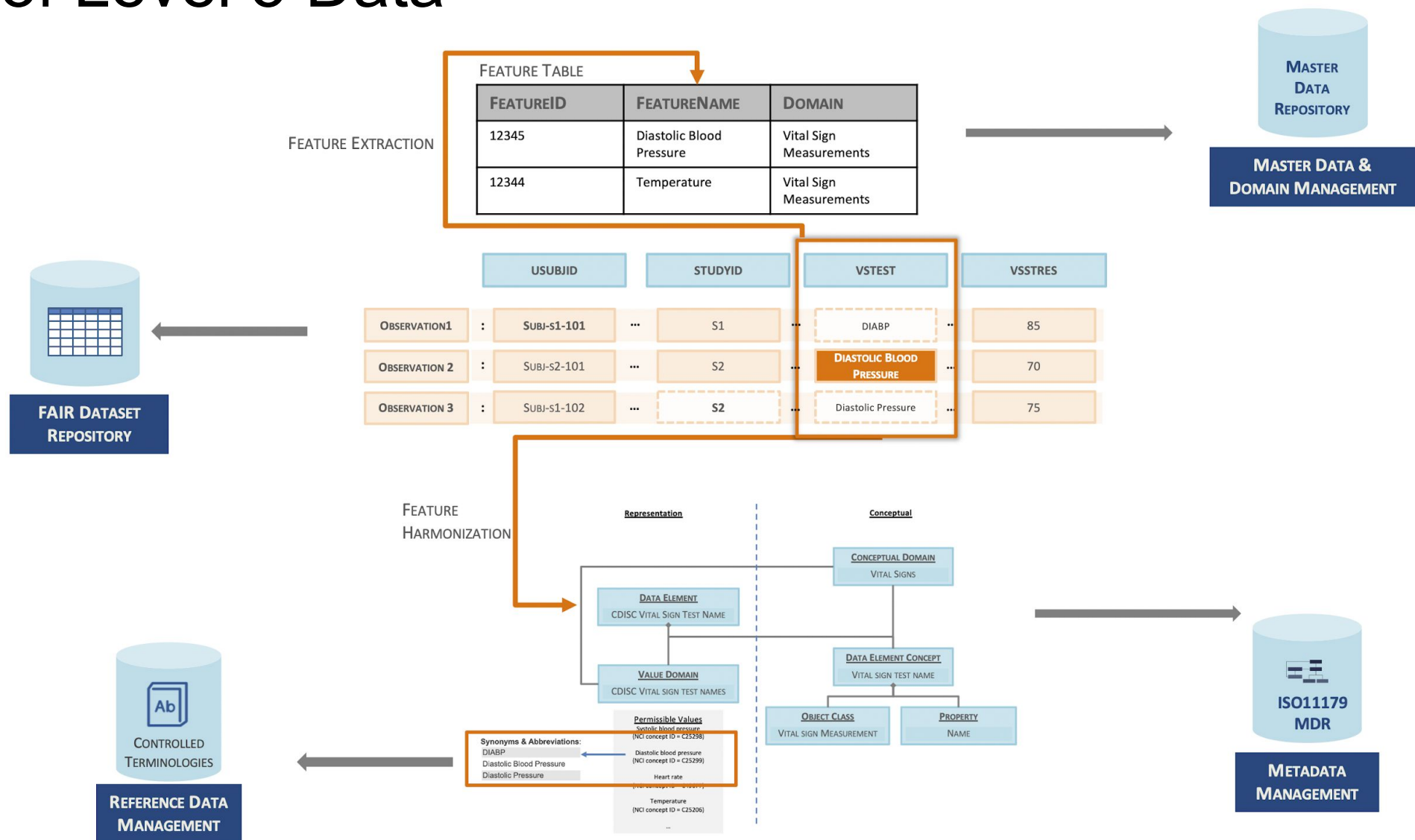
FAIR-DSM – Level 5 Use Case

As a data-driven enterprise, I want to be able to manage my research data assets for the purpose of supporting multiple researchers and implementation of common Data Governance policies.

Data Usage Goals: Data Asset Management, Data Re-annotation, Data Governance



Example of Level 5 Data



Using FAIR DSM

<https://fairdsm.biospeak.solutions/>

FAIR-DSM Assessment Result SummaryX

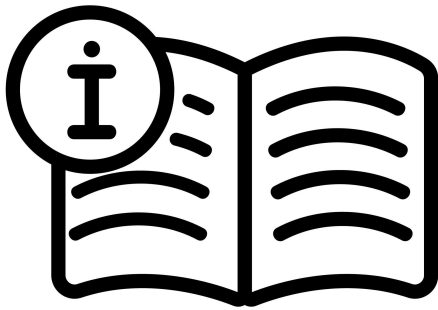
Based on the current assessment, the table below shows the percentage of indicators satisfied towards reaching each level of maturity.

Level	Representation & Format	Content & Context	Hosting Environment Capabilities	Overall Level % Completion	
Level 1	83%	75%	100%	86%	Details
Level 2	88%	44%	50%	62%	Details
Level 3	50%	0%	40%	30%	Details
Level 4	17%	0%	0%	7%	Details
Level 5	0%	0%	0%	0%	Details

[OK](#)[Print](#)

FAIR assessment hands-on activity

- ❑ RDA FAIR Data Maturity Model (<https://doi.org/10.15497/rda00050>)
- ❑ FAIR Checker (<https://fair-checker.france-bioinformatique.fr/>)
- ❑ FAIR Enough ([FAIR Enough](#))
- ❑ FAIR DataSet Maturity (DSM) (<https://fairplus.github.io/Data-Maturity/>,
<https://fairdsm.biospeak.solutions/>)
- ❑ “Do I-PASS for FAIR” ([Do I-PASS for FAIR. A self assessment tool to measure the FAIR-ness of an organization](#))



Working in pairs, pick one of the assessment tools or frameworks from the list and try to apply it to your data, project or resource.